Live Spectrometer Tool

Design and Implementation of a Reusable Measurement Interface

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1. Introduction and Motivation

In many laboratory environments, spectrometer measurements are performed using simple scripts that only display raw intensity values. While this approach is sufficient for quick checks, it is often not suitable for structured experiments, teaching demonstrations, or reproducible data acquisition.

The motivation of this project was to develop a reusable live spectrometer tool that can be used both in laboratory measurements and in teaching contexts. The focus was not only on displaying data, but also on providing real-time interaction, reference-based measurements, and a clear user interface that can be understood and used by students without modifying the source code.

Special emphasis was placed on usability and robustness, so that the tool can be reused in future lab courses and demonstrated during practical sessions.

2. System Overview

The software follows a simple and modular measurement pipeline:

Acquisition → Processing → Visualization → Saving

- Acquisition:
  Spectra are read via the Seabreeze library from the first available spectrometer. If no hardware is detected, a dummy spectrum is generated automatically, allowing development and testing without the device.
- Processing:
  The raw spectrum can be smoothed using a moving average filter. Dark and white reference measurements are supported to compute corrected transmission and absorbance.
- Visualization:
  Multiple spectra can be displayed simultaneously, and all important parameters can be adjusted interactively.
- Saving:
  Data can be saved manually or automatically, including reference measurements, in CSV format for further analysis.

3. User Interface Design

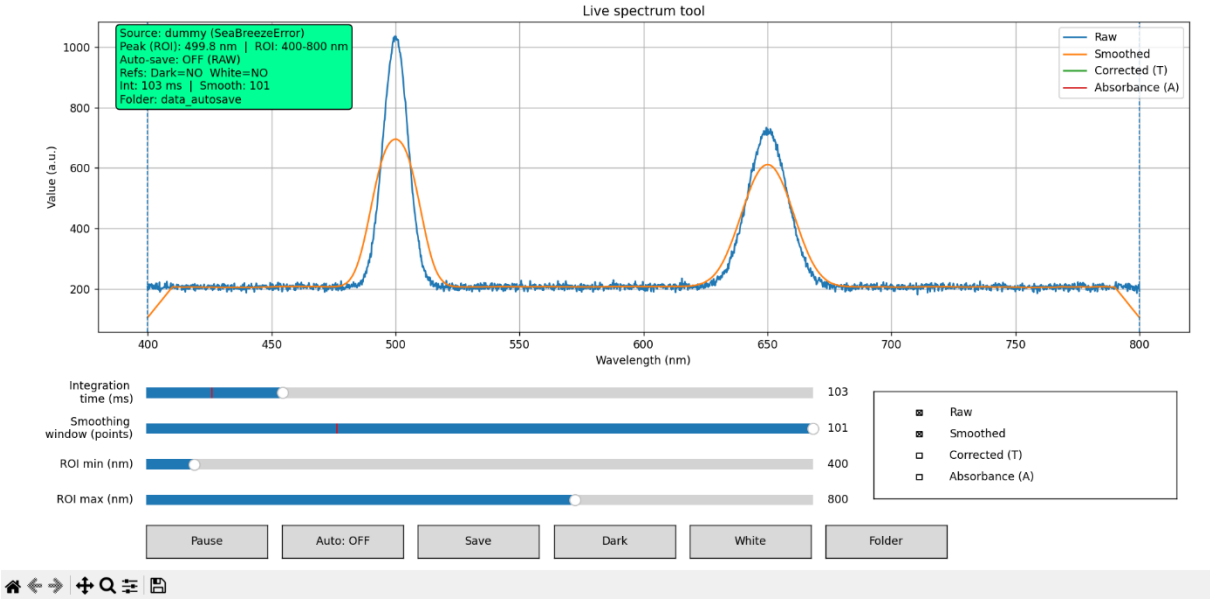

Figure 1: *Live spectrometer user interface showing raw, smoothed, corrected transmission and absorbance curves.*

The user interface was designed to be clear, compact, and suitable for teaching demonstrations. Multiple curves can be displayed simultaneously, including raw intensity, smoothed intensity, corrected transmission, and absorbance.

Interactive sliders allow the user to adjust:

- integration time (ms),
- smoothing window size,
- region of interest (ROI) for peak detection.

Checkboxes make it possible to enable or disable individual curves, avoiding visual overload. A status information box displays the current acquisition mode, peak wavelength, reference status, and active settings.

This layout allows students to immediately see how parameter changes affect the measured spectrum in real time.

4. Reference Measurements and Data Processing

Two reference measurements are supported:

- Dark reference: recorded with blocked light to capture sensor baseline and noise.
- White reference: recorded under a reference condition with maximum signal.

Using these references, the corrected transmission is calculated as:

$T = (raw - dark) / (white - dark)$

From the transmission, absorbance is derived as:

$A = -\log10(T)$

This approach allows relative measurements that are independent of absolute intensity values and is commonly used in optical experiments.

5. Discussion and Limitations

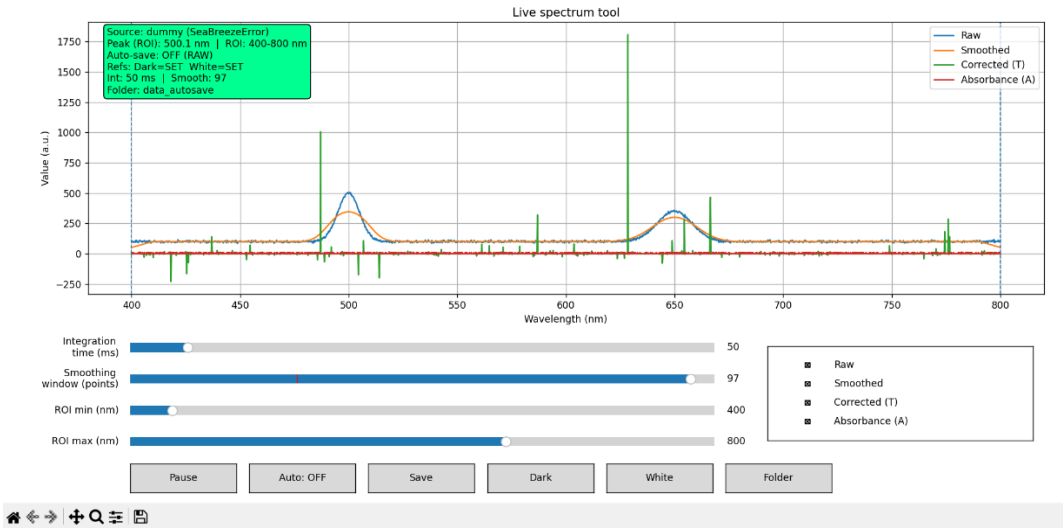

Figure 2: *Example of numerical instability in corrected transmission and absorbance caused by imperfect reference measurements.*

During testing, it was observed that reference-based correction can become numerically unstable if the dark and white references are very close to each other or contain significant noise. In such cases, large spikes may appear in the corrected transmission and absorbance curves.

This behavior is not a software error, but a direct consequence of the correction formula when the denominator *(white − dark)* becomes very small. The tool intentionally visualizes this effect, as it helps students understand the importance of proper reference acquisition and stable experimental conditions.

Therefore, the software does not hide such effects, but instead makes them visible as part of a realistic and educational measurement process.

6. Conclusion

A live spectrometer tool with an interactive graphical user interface was successfully implemented. The software supports real-time acquisition, reference-based correction, absorbance calculation, and reproducible data export.

The design focuses on usability and clarity, making the tool suitable not only for individual experiments but also for future lab courses and teaching demonstrations. The modular structure allows further extensions if required.